RISK ASSESSMENT SUMMARY SHEET	No.
Activity being risk assessed	**Research study (or student project study) involving human participants using virtual reality (VR) head-mounted display HMD.**

1. Details of the risk assessment.

Campus: Brayford	**Assessment Date: 24/03/25**
College / School / Department: School of Engineering and Physical Sciences	**Re-assessment Date: 24/03/26**
Location / Area: Multiple. Any lab space in which a specific area has been configured for exclusive use for the study. E.g. booth in INB1304.	**Risk assessment team members: Patrick Dickinson**
Accountable Manager: **Patrick Dickinson**	

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2. Details of further action necessary to control risk (with dates)

Task	Action	Who is Responsible	Date

3. Evaluation

This assessment is an accurate statement of the known hazards, risks and precautions. I certify that the control measures will prevent or, if this is not possible, control the risk subject to the level shown in section 4 (below) and that staff will be adequately trained and supervised, and the identified control measures implemented. The contents of this assessment will be communicated to staff and all relevant persons.

Signature of Assessor: PATRICK DICKINSON	Date: 24/03/25
Signature of Accountable Manager (if not Assessor):	Date:

4. Summary of risks with controls in place (indicate with 'X' where applicable).

Assessment of risk	Low	X	Medium Low		Medium High		High	

Risk to pregnant workers?	Yes	x	No	
Risk to disabled workers?	Yes	x	No	

Probability (likelihood):		Severity (Impact):	
1.	**Extremely Remote (Unlikely but Possible):** • A highly improbable situation, though theoretically possible.	1.	**Trivial or Negligible (Remote Impact):** • The hazard will not result in serious injury or illness, and there is a possibility of minor cuts or damage.
2.	**Possible (Occasional):** • An event with occasional likelihood, not a frequent occurrence but plausible.	2.	**Minor Injury (Potential Harm):** • The hazard may cause illness, injury, or equipment damage, but serious consequences are not expected. *Examples: Mild bruising, cuts, mild chemical irritation to eyes or skin; minor property damage.*
3.	**Probable (Anticipated):** • A scenario that is likely to occur within a given time frame, not surprising.	3.	**Serious Injury (Risk of Serious Consequences):** • The hazard can result in serious injury, illness, or property and equipment damage. *Examples: Loss of consciousness, burns, fractures, or injury resulting in absence from work for more than 3 days; serious property damage confined to the workroom or area.*
4.	**Likely or Frequent (Expected):** • A situation with a high probability of occurrence, expected to happen regularly.	4.	**Fatality, Major Injury (Imminent Danger):** • Imminent danger exists; the hazard is capable of causing death and widespread illness. *Examples: Death, property damage affecting the loss of one or more buildings.*

PROBABILITY	Risk Assessment Matrix 4 x 4				Risk Rating
	4	8	12	16	High
	3	6	9	12	Medium High
	2	4	6	8	Medium Low
	1	2	3	4	Low
SEVERITY					

Risk Rating and Recommended Actions:	
1 or 2 (Low):	Continue with existing controls but monitor for changes.
3 or 4 (Medium Low):	Requires attention to reduce the rating and regular ongoing monitoring.
6, 8, or 9 (Medium High):	Requires immediate attention to bring the risk down to an acceptable level.
12 or 16 (High):	Immediate action required. The risk is too high and not acceptable; stop immediately.

Risk Assessment Detail Sheet General hazards

Task	Hazard	Who might be harmed and how?	Initial risk (before controls)			Control Measures	Revised risk (after controls)		Overall risk
			P	S	PxS 3+		P	S	P x S
Using virtual reality (VR) Head-mounted display (HMD) with study participant(s) as part of lab study. Hazards apply to participant(s) and researcher(s) and should be managed by the researcher(s).	Trip, fall or collision with wall, furniture, or other people (including researcher(s) and other participants in multi-participant study).	Participant(s) Researcher(s)	2	2	4	Ensure VR floor area is free of objects or other trip hazards such as uneven floor. If participant is standing, ensure VR safety chaperone system is enabled and correctly mapped to floor space. If seated study, ensure HMD is only worn while seated, and that walls and objects not used in study are out of reach when seated. Participant should be briefed on HMD safety before using it, including how the chaperone system works. Participant should be made familiar with other study equipment (e.g. controllers, keyboard) before wearing HMD. Participants who may be vulnerable to fall/collision (e.g. pregnant) should either be screened out, or asked to self-assess individual risk and self-screen. To avoid collisions with people not involved with study, researcher should take reasonable measures ensure that other people do not accidentally enter the chaperoned VR floor space used the participant(s) during the study. For example, the floor area should not be located near a doorway through which someone might enter. The researcher should put up signs or otherwise warn other people that the experiment is being conducted. The researcher should observe the participant during the study, so that they can anticipate any collisions or other hazards that may arise. If the researcher is unable to do this (e.g. they are operating equipment) then an additional researcher should be present for this purpose. <u>Multiple participant studies</u>: If there are multiple participants using HMDs at the same time, then there should also be additional researchers present so that one researcher can exclusively observe one participant. Participants should not use the same or overlapping chaperoned floor areas. They should use different areas that are well separated (e.g. sperate booths). Studies that require participants to use the same physical space while wearing an HMD will need individual risk assessment. <u>Researcher(s):</u>	1	2	2

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			P	S	PxS 3+		P	S	P x S
						Generally, the researcher should avoid entering the floor space mapped for the chaperone system as much as possible while the participant is using the HMD, as the participant may not be able to see them. It is likely that the researcher will need to sometimes be in the chaperoned area to assist the participant or perform other ad-hoc functions (e.g. to help participant adjust or remove HMD). The researcher should define a protocol for such cases; for example, issuing a verbal warning, and communicate the protocol to the participant. If the researcher needs to enter the chaperoned floor space to perform some specific function as part of the study design, then similarly, an appropriate protocol should be defined and adopted to minimise risk of collision.			
	HMD wire entanglement	Participant(s)	3	1	3	If HMD is connected to other equipment by a wire, then participant should be made aware of possible trip/entanglement hazard and given advice on what to do if entangled (e.g. ask for help).	2	1	2
	Experiencing uncomfortable levels of VR Sickness/ eye strain	Participant(s)	2	2	4	Probability is dependant on specific VR experience, and participant's individual susceptibility. Participants should be warned about the possibility of sickness and eye strain and told that they can stop the study if they are experiencing significant symptoms. Researcher should show the participant how to fit the HMD correctly (including adjustment of IPD settings, if available) to minimise risk. In studies which are investigating sickness symptoms (e.g. designed to elicit some level of sickness symptoms), participants should be informed that this is the case.	2	1	2
	Other hazards identified by manufacturer	Participant(s)	2	2	4	Researcher should familiarise themselves with any manufacturer H&S guidelines. For example: Meta Quest 2: https://www.meta.com/gb/quest/safety-center/quest-2/ Meta Quest Pro: https://www.meta.com/gb/quest/safety-center/quest-pro/ Meta Quest 3: https://www.meta.com/gb/quest/safety-center/quest-3/ Meta Quest 3S: https://www.meta.com/gb/quest/safety-center/quest-3s/	2	1	2

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			P	S	PxS 3+		P	S	P x S
	Fatigue	Participant(s)	2	1	2	Researcher should consider whether participant's time spent using VR HMD is reasonable, paying attention to any manufacturer guidelines. Researcher could consider giving participants breaks if appropriate.	1	1	1
	Physical discomfort	Participant(s)	2	1	2	VR equipment can sometimes be uncomfortable, especially if not fitted properly. This may cause significant discomfort to participants. Researcher should show the participant how to fit the HMD correctly (including adjustment of IPD settings, if available) to minimise risk. Participants should be told that they can ask for help resolving minor issues and can stop the study if they are experiencing significant discomfort.	1	1	1
	Psychological discomfort	Participant(s)	2	2	4	Researcher should be aware that experiences in VR can be considerably more emotionally and psychologically impactful than watching them on a screen. If the content has potential for negative impacts on the participant, then the researcher should ensure that the participant is aware of the content, and that it may be more impactful than expected in VR, when obtaining consent. This is particularly relevant for participants who have not used VR before. Researcher should emphasise that participant can withdraw before or during the study, if they have concerns about the content. For some studies, researchers may consider using participants who have a minimum level of experience of using VR.	1	2	2
	Participant is unable to stop the study quickly	Participant(s)	2	2	4	Participants may wish to stop the study while they are using the HMD, for a number of reasons (e.g. sickness, physical or emotional discomfort), but not know how to indicate this or seek help from the researcher. As a result, they may be unnecessarily exposed to additional duress. To avoid this, the researcher should define a protocol to enable participant to quickly indicate to the researcher that they wish to stop the study (e.g. raising hand or verbalising), and to enable the researcher to help them exit the study as quickly and safely as possible (e.g. assist with removing HMD). This protocol should be communicated to participants before starting the study.	1	2	2

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			P	S	PxS 3+		P	S	P x S
	Additional hazards for participants with sensory or motor impairments e.g. wheelchair users.	Participant(s)	2	2	4	Researcher should take time to ensure that the study condition is accessible for individual needs of participant. Researcher should pay particular attention to risk of fall/collision and be prepared for a more detailed conversation about safety, if appropriate. Participants should be asked to self-assess individual risk, according to their specific needs, and self-screen if necessary. Researcher should pay extra attention to possible cable entanglement (wired HMDs) and be prepared to assist if required.	1	2	2